

Sneha Agarwal

Software Engineer | Backend / Full-Stack Developer (MERN)

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Education

B.Tech in Computer Science and Engineering

University of Calcutta | 2023 – 2027 (Expected) | **CGPA: 7.32**

Senior Secondary (Class XII) - CBSE | 2023 | **75%**

Secondary (Class X) - ICSE | 2021 | **85%**

Projects

Money Mentor

Tech Stack: Next.js, React, Node.js, Express, REST APIs, SQL, Render

Architected a full-stack financial tracking platform using **Next.js (SSR)** and **Node.js backend services**, enabling secure expense ingestion, budgeting workflows, and **behavioral insights**.

Built scalable **relational data models**, optimized **asynchronous service pipelines**, and deployed using **cloud-based hosting on Render** to ensure low-latency performance and production stability.

College Admission Management Portal

Tech Stack: Node.js, Express, MySQL, HTTP-based APIs, Razorpay, PDFKit

Engineered a **role-aware admissions system** processing 1000+ applications, leveraging **Express-based middleware**, **authorization policies**, and **indexed relational queries** achieving sub-2s response times.

Integrated **Razorpay payment processing**, automated **PDF document generation and verification**, and delivered **data visualization modules** to reduce manual effort and improve operational transparency.

Scientific Publication Management System

Tech Stack: Node.js, Express, MySQL, React, JWT, bcrypt

Developed a secure backend with **token-based authentication**, **credential encryption**, and **policy-driven access control** supporting multi-role publication workflows.

Designed **3NF-compliant relational schemas** and **service-oriented API modules** to manage CRUD pipelines, automate account lifecycle management, and generate **reporting and metrics insights**.

Technical Proficiencies

- Languages: C++, Java, Python, JavaScript
- Backend: Node.js, Express.js, RESTful APIs, Authentication, Authorization
- Databases: MySQL, MongoDB, Relational Database Design
- Frontend: React.js, Next.js, HTML5, CSS3
- Tools and Platforms: Git, GitHub, Postman, VS Code, Render, Vercel

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks

Achievements

- **Runner-up**, Hult Prize OnCampus 2023–24, University of Calcutta, for a social entrepreneurship solution.
- **2nd Position**, Start-Up Hackathon organized by The Idea Company.
- **Selected Participant**, Smart India Hackathon 2025 (Internal Round), Team Hack4Change.
- Certificate of **Excellence**, CodeRush Weekly, for consistent programming and problem-solving performance.

Languages

English, Hindi, Bengali